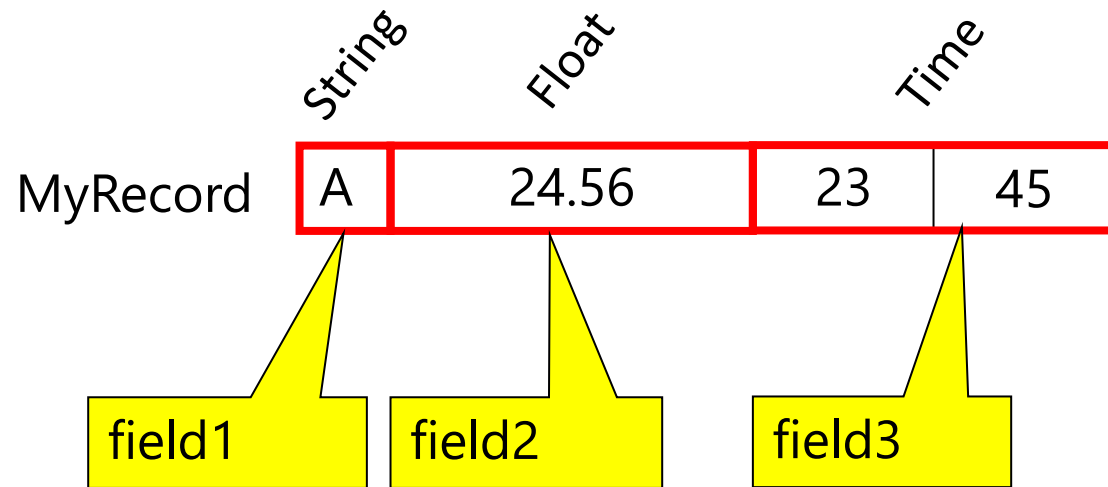


```
print("Computer Programming 2026\n")  
print("Department of Geography\n")  
print("7 May 2026")  
  
print("Week 11")
```

DATAFRAMES

A new data structure

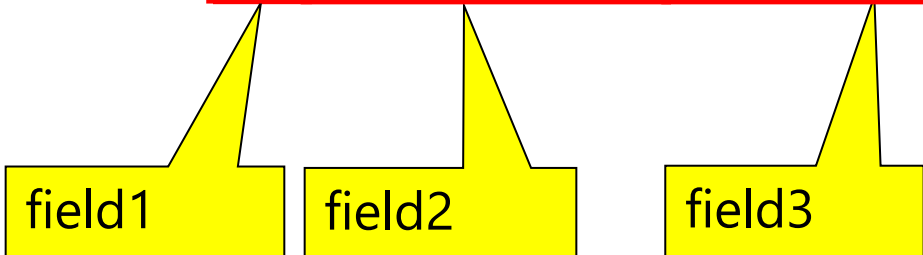
- Tabular data structure of mixed data types:



A new data structure

- Tabular data structure of mixed data types:

	String	Float	Time	
MyRecord	A	24.56	23	45
YourRecord	B	27.12	22	30



The diagram illustrates the mapping of field names to table columns. Three yellow callout boxes labeled 'field1', 'field2', and 'field3' are positioned below the table. Yellow lines connect 'field1' to the 'String' column, 'field2' to the 'Float' column, and 'field3' to the first column of the 'Time' header (which corresponds to the first value in the time pairs, e.g., 23 and 22).

A new data structure

- Tabular data structure of mixed data types:

	String	Float	Time	
MyRecord	A	24.56	23	45
YourRecord	B	27.12	22	30
HerRecord	C	29.56	21	15

field1

field2

field3

Pandas Library



What is Pandas?

- Pandas is a Python library used for working with datasets.
- It has functions for analyzing, cleaning, exploring, and manipulating data.
- The name "Pandas" has a reference to both "Panel Data", and "Python Data Analysis" and was created by Wes McKinney in 2008.

Why use Pandas?

- Pandas allows us to analyze big data and make conclusions based on statistical theories.
- Pandas can clean messy datasets, and make them readable and relevant.
- Pandas is a key tool in data science.

Pandas DataFrame



A **Pandas DataFrame** is a two-dimensional, size-mutable, potentially heterogeneous, tabular data structure with labeled axes.

A Pandas DataFrame consists of three main components: **data**, **rows**, and **columns**.

The diagram illustrates the structure of a Pandas DataFrame. It features a table with 7 rows and 6 columns. The columns are labeled 'Name', 'Team', 'Number', 'Position', and 'Age'. The rows are indexed from 0 to 6. A blue arrow labeled 'Columns' points to the column headers. An orange arrow labeled 'Rows' points to the row indices. A pink arrow labeled 'Data' points to the data cells of the table. The data cells are highlighted with a pink border. The table is enclosed in a green rounded rectangle with a green border. In the bottom right corner of the rectangle is a green logo consisting of two interlocking 'G' shapes.

	<i>Name</i>	<i>Team</i>	<i>Number</i>	<i>Position</i>	<i>Age</i>
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

Pandas DataFrame



Handling data in a Pandas DataFrame:

- Creating a DataFrame
- Dealing with rows and columns
- Indexing and selecting data
- Working with missing data

Exercise (betting system)

- The program should store in a tabular structure:
 - date of the game
 - name of the home team
 - name of the away team
 - bet result (1 X 2)
- After inserting a few entries, ask the user to select all entries of a specific day and display them on screen.

Exercise (betting system)

- The program should store in a tabular structure:
 - date of the game
 - name of the home team
 - name of the away team
 - bet result (1 X 2)

```
import pandas as pd
```

```
bet = pd.DataFrame(columns = ["Date", "Home Team", "Away Team", "Bet Result"])
```

Exercise (betting system)

```
import pandas as pd

bet = pd.DataFrame(columns = ["Date", "Home Team", "Away Team", "Bet Result"])

while True:
    question = input("Do you want to bet on a new game? (y/n) ")
    if question == "y":
        date = input("Insert the date of the game (mm/dd/yyyy): ")
        HT = input("Insert the name of the home team: ")
        AT = input("Insert the name of the away team: ")
        result = input("Insert your bet (1 X 2): ")
        bet.loc[len(bet)] = [pd.to_datetime(date), HT, AT, result]
    else:
        break
```

Exercise (betting system)

- After inserting a few entries, ask the user to select all entries of a specific day and display them on screen.

```
query = input("Which day do you want to select? (mm/dd/yyyy) ")  
query_output = bet[bet["Date"] == pd.to_datetime(query)]
```

Exercise (betting system)

- After inserting a few entries, ask the user to select all entries of a specific day and display them on screen.

```
query = input("Which day do you want to select? (mm/dd/yyyy) ")
query_output = bet[bet["Date"] == pd.to_datetime(query)]

if len(query_output) != 0:
    print(f"\nYour bets in day {query} are:")
    display(query_output)
else:
    print("\nYou did not bet any game in day", query)
```

Let's continue on Jupyter Notebook...